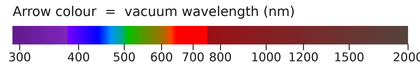
Americium

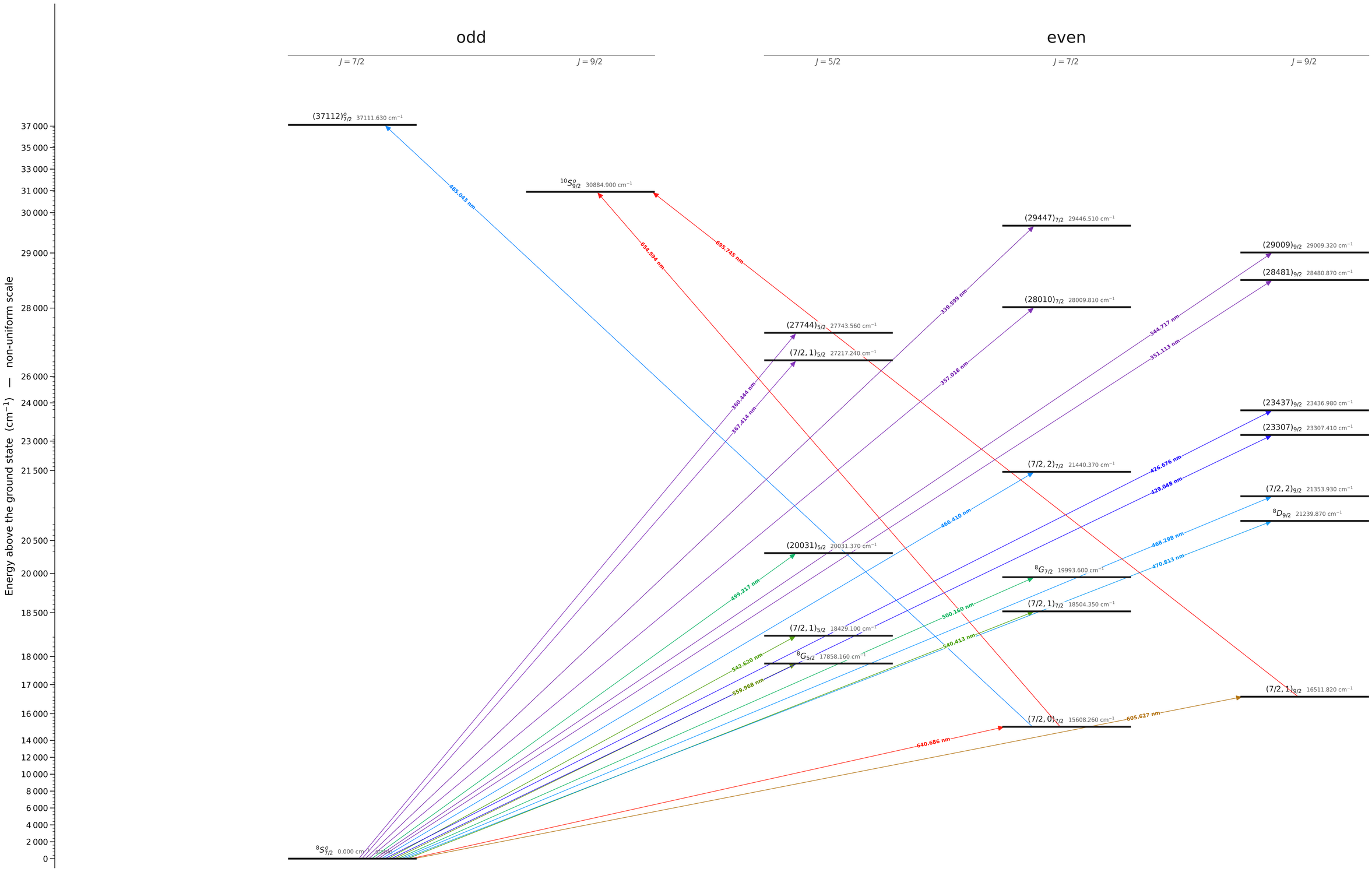
Am (Am I). Classified transitions below 2 μm (levels and lines from the literature): 21 levels, 21 lines.

Am^+ ionisation limit 48182.00 cm^{-1} = 5.97381 eV (207.546 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|\langle J \| er \| J' \rangle|$ in ea_0
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.
Level caption: energy (from the cited literature; NIST lists no excited levels) and measured lifetime (where one exists).

Arrow colour = vacuum wavelength (nm)





Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$\lvert\langle J \| er \| J' \rangle\rvert$ (ea_0)	A (s^{-1})	Γ/2π up. (MHz)	branch (%)	Use
$^8S°_{7/2}$ – $(23437)_{9/2}$	426.6761	426.5561	702.6230	–	–	–	–	first step of the Mainz two-step resonance-ionization sch…
$^8S°_{7/2}$ – $(7/2, 1)_{9/2}$	605.6268	605.4591	495.0119	–	–	–	–	605.46 nm line

λ, ν from the level energies of the cited literature (NIST lists no excited levels) unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

^{241}Am hyperfine structure (I = 5/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
$^8S°_{7/2}$	-17.144(8)	123.82(10)	F=1: +259.20 F=2: +189.54 F=3: +98.31 F=4: +1.43 F=5: -79.87 F=6: -119.06
$(7/2, 0)_{7/2}$	1415(15)	1139(300)	F=1: -15308.46 F=2: -12803.95 F=3: -8925.12 F=4: -3525.51 F=5: +3590.18 F=6: +12666.05
$(7/2, 1)_{9/2}$	1715(15)	-2968(300)	F=2: -24938.79 F=3: -18904.02 F=4: -11203.91 F=5: -2135.26 F=6: +7930.94 F=7: +18549.53
$^8G_{5/2}$	-561(15)	-30(300)	F=0: +4884.25 F=1: +4330.85 F=2: +3221.35 F=3: +1550.35 F=4: -690.25 F=5: -3511.25
$(7/2, 1)_{5/2}$	-1271(15)	-90(300)	F=0: +11059.19 F=1: +9809.67 F=2: +7302.53 F=3: +3521.60 F=4: -1557.41 F=5: -7966.85
$(7/2, 1)_{7/2}$	-959(15)	884(30)	F=1: +11265.91 F=2: +9094.62 F=3: +5932.45 F=4: +1893.11 F=5: -2871.81 F=6: -8172.77
$^8G_{7/2}$	183(15)	420(300)	F=1: -1832.79 F=2: -1586.90 F=3: -1173.10 F=4: -537.43 F=5: +392.05 F=6: +1705.30
$(20031)_{5/2}$	468(15)	528(30)	F=0: -3723.05 F=1: -3381.98 F=2: -2652.34 F=3: -1439.18 F=4: +399.96 F=5: +3055.03
$^8D_{9/2}$	561(15)	2668(30)	F=2: -6485.32 F=3: -5603.98 F=4: -4117.57 F=5: -1759.28 F=6: +1804.44 F=7: +6973.80
$(7/2, 2)_{9/2}$	1103(15)	1289(300)	F=2: -14578.16 F=3: -11655.29 F=4: -7607.74 F=5: -2306.59 F=6: +4409.30 F=7: +12733.27
$(7/2, 2)_{7/2}$	1268(15)	-1499(30)	F=1: -15069.16 F=2: -12104.67 F=3: -7818.55 F=4: -2403.53 F=5: +3883.44 F=6: +10721.12
$(23307)_{9/2}$	-198(15)	-1559(30)	F=2: +2006.63 F=3: +1880.60 F=4: +1530.69 F=5: +801.00 F=6: -503.31 F=7: -2616.10
$(23437)_{9/2}$	-66(15)	-2968(300)	F=2: -452.79 F=3: +239.58 F=4: +816.49 F=5: +981.14 F=6: +362.55 F=7: -1484.47
$(7/2, 1)_{5/2}$	327(15)	630(30)	F=0: -2418.78 F=1: -2243.08 F=2: -1835.03 F=3: -1081.28 F=4: +188.14 F=5: +2199.90
$(27744)_{5/2}$	219(15)	1859(300)	F=0: -613.41 F=1: -840.70 F=2: -1127.99 F=3: -1140.72 F=4: -377.05 F=5: +1832.17
$(28010)_{7/2}$	-63(15)	-630(300)	F=1: +371.46 F=2: +425.35 F=3: +438.72 F=4: +330.63 F=5: -6.86 F=6: -708.65
$(28481)_{9/2}$	-126(15)	480(300)	F=2: +1950.99 F=3: +1429.38 F=4: +789.86 F=5: +80.41 F=6: -639.01 F=7: -1296.45
$(29009)_{9/2}$	192(15)	1169(300)	F=2: -2102.74 F=3: -1877.80 F=4: -1441.47 F=5: -676.84 F=6: +562.25 F=7: +2451.18
$(29447)_{7/2}$	1325(15)	-899(300)	F=1: -15389.20 F=2: -12482.02 F=3: -8217.63 F=4: -2711.66 F=5: +3881.72 F=6: +11369.77
$^{10}S°_{9/2}$	2230(15)	150(300)	F=2: -30600.67 F=3: -23954.14 F=4: -15074.61 F=5: -3947.10 F=6: +9447.15 F=7: +25130.60
$(37112)°_{7/2}$	2188(15)	-450(300)	F=1: -24861.54 F=2: -20356.05 F=3: -13646.00 F=4: -4789.22 F=5: +6137.22 F=6: +19036.95

Measured A, B constants; sources in the data file.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
$^8S°_{7/2}$ – $(29447)_{7/2}$	339.599	243-241	-3598.00
$^8S°_{7/2}$ – $(29009)_{9/2}$	344.717	243-241	5606.00
$^8S°_{7/2}$ – $(28481)_{9/2}$	351.113	243-241	-420.00
$^8S°_{7/2}$ – $(28010)_{7/2}$	357.018	243-241	-2338.00
$^8S°_{7/2}$ – $(7/2, 1)_{5/2}$	367.414	243-241	-2698.00
$^8S°_{7/2}$ – $(23437)_{9/2}$	426.676	243-241	3837.00
$^8S°_{7/2}$ – $(23307)_{9/2}$	429.048	243-241	3478.00
$(7/2, 0)_{7/2}$ – $(37112)°_{7/2}$	465.043	243-241	450.00
$^8S°_{7/2}$ – $(7/2, 2)_{7/2}$	466.410	243-241	-600.00
$^8S°_{7/2}$ – $(7/2, 2)_{9/2}$	468.298	243-241	-2009.00
$^8S°_{7/2}$ – $^8D_{9/2}$	470.813	243-241	3867.00
$^8S°_{7/2}$ – $(20031)_{5/2}$	499.217	243-241	1709.00
$^8S°_{7/2}$ – $^8G_{7/2}$	500.160	243-241	7075.00
$^8S°_{7/2}$ – $(7/2, 1)_{7/2}$	540.413	243-241	-5276.00
$^8S°_{7/2}$ – $(7/2, 1)_{5/2}$	542.620	243-241	-1289.00
$^8S°_{7/2}$ – $^8G_{5/2}$	559.968	243-241	4167.00
$^8S°_{7/2}$ – $(7/2, 1)_{9/2}$	605.627	243-241	-6745.00
$^8S°_{7/2}$ – $(7/2, 0)_{7/2}$	640.686	243-241	-7015.00
$(7/2, 0)_{7/2}$ – $^{10}S°_{9/2}$	654.594	243-241	1799.00
$(7/2, 1)_{9/2}$ – $^{10}S°_{9/2}$	695.745	243-241	1529.00

Shift = ν(first isotope) − ν(second isotope).